

HUMZA KHAN

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EDUCATION

Northwestern University

Expected May 2026

B.S. Industrial Engineering and Management Sciences; Economics

GPA: 3.97/4.00

Relevant Coursework: Probability (grader and student), Statistics, Data Science, Applied Linear Algebra, Microeconomics, Macroeconomics, Data Management and Processing, Foundations of Optimization, Econometrics

Awards and Societies: Dean's List (5x), Tau Beta Pi

WORK EXPERIENCE

P33 Chicago, Deep Tech Intern

June 2024 – August 2024

- Researched the national and regional quantum ecosystem focusing on technology commercialization and models designed to accelerate the development of quantum, drawing on historical evidence from related hard-tech industries
- Developed regional quantum and advanced materials industry maps using large-scale open-source databases such as OpenAlex and the U.S. Patent Office
- Built a live grants dashboard for all funding opportunities related to CHIPS, IRA and IIJA to harmonize various tracking sources

U.S. Department of State, Technology Policy Intern

September 2023 – May 2024

- Mapped global semiconductor R&D capabilities, encompassing basic research, physical infrastructure, applications, and end users, with a focus on geopolitical, innovation, and economic implications.
- Employed the US Patent Office API and a custom-built web scraper to generate semiconductor ecosystem analytics at the country and institutional level from 250,000+ IEEE publications and 1.2 million patents
- Utilized data visualization packages to aggregate and present large dataset analytics for the semiconductor ecosystem
- Designed custom databases and data templates using SQL and Excel to improve data standardization and visualization

RailTEC at the University of Illinois Urbana-Champaign, Research Assistant

May 2023 – August 2023

- Carried out data analysis and modeling to understand the causes of catastrophic rail accidents and support the formulation of better rail safety policy
- Utilized scraping algorithms to analyze Federal Railroad Administration and CHEMTREC datasets to identify and classify serious hazardous tank-car accidents

Haylon Technologies, Student Consultant

September 2022 – May 2023

- Conducted market research on a variety of sectors including energy, batteries, and mining
- Researched product analogs and vetted potential investors to provide strategy and funding recommendations
- Developed SPICE and industry-standard PCB software models of battery components and cells, including a lithium-ion battery to within 5% accuracy of all characteristics, custom-modeled components, and overall cell integration

PROJECTS & PUBLICATIONS

Power Up; Stanford Economic Review: This commentary examines regulatory and policy-driven challenges facing U.S. energy infrastructure, with targeted policy reforms aimed at strengthening grid resilience and accelerating modernization.

Making Waves; P33 Chicago: This report provides recommendations for key industry and governmental stakeholders in the national and regional quantum ecosystem, focusing on the areas of talent acquisition, technology commercialization and industry resilience to macroeconomic shocks

International Semiconductor R&D Capabilities Mapping; U.S. Department of State: This interactive dashboard and report provides in-depth analyses of the global semiconductor ecosystem including specific facilities, research areas and international collaborations

SKILLS & CERTIFICATES

Data Analysis and Visualization: Python (Udemy Certification), R, SQL, Tableau

Engineering Software: LTSpice, Altium, AMPL, Gurobi

Research: Four years of competitive high school policy debate, produced over 1000 pages of research and analysis